Beechcraft A36 Bonanza (A36)

Continental IO-550-B, 300 hp · 6 seats · Typical A36 (1984+, IO-550-B, 3,650 lb) POH procedures; earlier IO-520-BB aircraft and pre-1984 quadrant / gear-flap switch layouts differ.

V-speeds & limits

Vr	**71 KIAS (liftoff)**
Vx	**84 KIAS**
Vy	**100 KIAS**
Best glide	**110 KIAS (gear & flaps up, prop low RPM)**
Vfe	**152 KIAS (approach 15°) / 123 KIAS (full 30°)**
Vlo / Vle	**154 KIAS**
Va	**141 KIAS (3,650 lb)**
Vno	**167 KIAS**
Vne	**205 KIAS**
Vs / Vso	**approx 64 / 52 KIAS (max weight)**
Vref	**75–80 KIAS final (full flaps; 70 KIAS short field)**

Normal procedures

Before Start

Preflight inspection	**COMPLETE**
Passenger briefing	**COMPLETE (belts, doors, exits, no smoking)**
Seats, belts, shoulder harnesses	**ADJUST & LOCK**
Cabin door / utility doors	**CLOSED & LATCHED**
Parking brake	**SET**
Avionics master	**OFF**
Landing gear switch	**DOWN (identify – gear switch and flap switch sit on opposite sides of the quadrant)**
Throttle / propeller / mixture	**IDENTIFY on quadrant (throttle–prop–mixture, left to right on 1984+)**
Fuel selector	**FULLEST TANK (LEFT or RIGHT – there is no BOTH)**
Fuel quantity	**CHECK both gauges**
Cowl flaps	**OPEN**
Alternate static source	**OFF (normal)**
Battery & alternator switches	**ON**
Landing gear indicator	**GREEN (down)**
Annunciators / warning lights	**TEST**
Beacon	**ON**

Engine Start

Propeller area	**CLEAR**
Mixture	**FULL RICH**
Propeller	**HIGH RPM (full forward)**
Throttle	**FULL OPEN**
Auxiliary fuel pump	**HI until fuel flow peaks (3–5 sec), then OFF**
Throttle	**1/4 OPEN (approx 1/4 in)**

▶ Engine hot or flooded?

No:

Yes:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Mixture	**ADVANCE to FULL RICH as engine fires, throttle to idle**
Ignition / starter	**START (release when engine fires)**
Throttle	**1000–1200 RPM**
Oil pressure	**30 psi within 30 sec (normal 30–60 psi)**
Alternator	**CHECK charging (loadmeter / voltmeter)**
Avionics master	**ON**
Engine instruments	**CHECK**

Before Taxi

Brakes	**CHECK on initial roll**
Flight instruments	**CHECK & SET (altimeter, HSI / DG, turn coordinator during turns)**
Avionics / GPS	**SET**
Autopilot / electric trim	**CHECK & OFF**
Elevator trim	**SET for takeoff**
Fuel selector	**CHECK fullest tank (LEFT or RIGHT)**
Cowl flaps	**OPEN**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Mixture	**FULL RICH (lean per placard above approx 5,000 ft elevation)**
Throttle	**1700 RPM**
Magnetos	**CHECK L & R (150 RPM max drop, 50 RPM max difference)**
Propeller	**CYCLE 2–3 times (RPM drops, then returns)**
Engine instruments	**CHECK (oil pressure 30–60 psi, oil temp in green, fuel flow steady)**
Alternator	**CHECK charging**
Suction / gyro pressure	**CHECK in green**
Throttle	**IDLE check, then 1000 RPM**
Fuel selector	**FULLEST TANK (feel for detent)**
Auxiliary fuel pump	**OFF**
Cowl flaps	**OPEN**
Propeller	**HIGH RPM**
Flaps	**UP (check indicator)**
Trim	**SET takeoff range**
Landing gear switch	**DOWN – IDENTIFY (hand on gear switch, not flap switch)**
Pitot heat	**AS REQUIRED**
Lights	**AS REQUIRED**
Transponder	**ALT**
Doors & windows	**LATCHED**
Takeoff briefing	**COMPLETE (abort point; gear up only with positive rate and no usable runway)**

After Takeoff / Climb

Landing gear	**UP (positive rate, no usable runway, below 154 KIAS)**
Gear indicator lights	**OUT (gear up and locked)**
Flaps	**UP**
Climb power	**25 in Hg / 2500 RPM (cruise climb) – full throttle / 2700 RPM permitted**
Mixture	**LEAN per POH climb fuel-flow schedule above approx 5,000 ft**
Cowl flaps	**OPEN (CHT / oil temp in green)**
Auxiliary fuel pump	**OFF (LO if fuel flow fluctuates)**
Airspeed	**110–120 KIAS (cruise climb)**
Engine instruments	**CHECK**

Cruise

Cruise power	**SET (2300–2500 RPM, 23–25 in Hg typical)**
Mixture	**LEAN per POH cruise table / EGT**
Cowl flaps	**CLOSED (as required)**
Fuel selector	**ALTERNATE LEFT / RIGHT to keep tanks balanced – note time**
Fuel quantity / flow	**CHECK against plan**
Engine instruments	**CHECK (oil temp 240°F max)**
Altimeter / heading	**CHECK**

Descent

Altimeter	**SET**
Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Mixture	**ENRICH progressively**
Power	**REDUCE gradually (avoid rapid cooling)**
Cowl flaps	**CLOSED**
Airspeed	**PLAN below 154 KIAS before gear extension**
Approach briefing	**COMPLETE**
Seat belts / harnesses	**SECURE**

Before Landing

Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Auxiliary fuel pump	**OFF**
Mixture	**FULL RICH**
Cowl flaps	**AS REQUIRED**
Airspeed	**BELOW 154 KIAS**
Landing gear	**DOWN – GREEN light (identify gear switch, not flap switch)**
Flaps	**APPROACH below 152 KIAS; DOWN on final below 123 KIAS**
Propeller	**HIGH RPM (on final)**
Airspeed	**75–80 KIAS final (70 KIAS short field)**
Landing light	**ON**
Landing gear	**GREEN – CONFIRMED on final**

After Landing

Flaps	**UP (look and identify the flap switch before moving)**
Cowl flaps	**OPEN**
Auxiliary fuel pump	**OFF**
Pitot heat	**OFF**
Landing light / strobes	**AS REQUIRED**
Transponder	**STBY / as required**
Trim	**SET takeoff**

Shutdown

Parking brake	**SET**
Throttle	**1000 RPM (cool-down)**
Avionics master	**OFF**
Electrical equipment / lights	**OFF**
Mixture	**IDLE CUTOFF**
Ignition	**OFF (after propeller stops)**
Battery & alternator switches	**OFF**
Control lock	**INSTALL**
Chocks / tiedowns	**AS REQUIRED**

Abnormal procedures

Alternator Failure / Low Voltage

ALT warning light, loadmeter zero or voltage below 24 V

Alternator switch ..... **OFF, then ON (reset)**

Alternator field circuit breaker ..... **CHECK / RESET once**

▶ Alternator restored?

Yes:

Loadmeter / voltmeter ..... **MONITOR**

No:

Alternator switch ..... **OFF**

Non-essential electrical load ..... **REDUCE (battery only, limited endurance)**

Landing gear ..... **PLAN early extension while battery power remains (hand crank if needed)**

Deferred: Alternator failed – battery power only; extend gear early, land as soon as practical. ..... **NOTED**

Low Oil Pressure

Oil pressure below 30 psi in flight

Oil pressure / oil temperature ..... **CHECK and cross-check**

▶ Oil temperature rising or pressure below 10 psi?

Yes – genuine:

Power ..... **REDUCE to minimum required**

Airport ..... **DIVERT to nearest suitable**

Engine ..... **EXPECT FAILURE – stay within glide of a landing area**

Deferred: Low oil pressure – land as soon as possible, expect possible engine failure. ..... **NOTED**

No – gauge suspect:

Flight ..... **CONTINUE, monitor oil temperature and pressure**

Deferred: Oil pressure indication suspect – maintenance check required. ..... **NOTED**

High Oil / Cylinder Head Temperature

Cowl flaps ..... **OPEN**

Mixture ..... **ENRICH**

Airspeed ..... **INCREASE (lower pitch attitude)**

Power ..... **REDUCE**

▶ Temperature returning to green?

Yes:

Flight ..... **CONTINUE, monitor**

No:

Oil pressure ..... **MONITOR for drop**

Deferred: High engine temperature – land as soon as practical, cooling / baffling check required. ..... **NOTED**

Rough Running / Fuel Flow Fluctuation

Auxiliary fuel pump ..... **LO (HI only if fuel flow very low – may flood engine at low power)**

Mixture ..... **ADJUST for smooth running**

Fuel selector ..... **SWITCH TANKS**

Magnetos ..... **CHECK L / R individually, then BOTH**

Engine instruments ..... **CHECK (CHT, EGT, fuel flow)**

▶ Engine running smoothly?

Yes:

Deferred: Rough running cleared – leave aux pump LO if used; report to maintenance. ..... **NOTED**

No:

Power ..... **REDUCE to smoothest setting**

Deferred: Engine rough – land as soon as practical at nearest suitable airport. ..... **NOTED**

Flap Failure / Asymmetry

Flap switch ..... **NEUTRAL / stop – do not continue extension**

Flap circuit breaker ..... **CHECK / RESET once**

▶ Flaps operable and symmetric?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Landing ..... **AT CURRENT FLAP SETTING – add 10 KIAS to approach speed, longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, increased landing distance. ..... **NOTED**

Landing Gear Fails to Retract / Extend Normally

Airspeed ..... **BELOW 154 KIAS**

Landing gear circuit breaker ..... **CHECK / RESET once**

Landing gear switch ..... **CYCLE once to the desired position**

▶ Gear in desired position?

Yes:

Gear indication ..... **CONFIRM**

Deferred: Gear system fault – do not retract after landing; maintenance inspection required. ..... **NOTED**

No – failed to retract:

Landing gear switch ..... **DOWN – leave down**

Airspeed ..... **BELOW 154 KIAS**

Deferred: Gear will not retract – gear down, reduced performance and range, land as soon as practical. ..... **NOTED**

No – failed to extend:

Use Emergency Gear Extension (Hand Crank).

Cabin Door Open in Flight

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 110 KIAS**

Belts / loose items ..... **SECURE**

Do not attempt to close the door by reaching across while flying the airplane. Expect increased noise and some buffet.

Deferred: Cabin door unlatched – return and land as soon as practical, normal approach speed. ..... **NOTED**

Vacuum / Attitude Indicator Failure

Suction gauge	**CHECK**
Attitude / heading indicators	**CROSS-CHECK with turn coordinator, compass, GPS, standby instruments**
Failed instruments	**COVER**
Autopilot	**OFF if coupled to failed gyro**
IMC	**EXIT / AVOID**
Deferred: Vacuum / gyro failure – partial panel; prefer VMC approach.	**NOTED**

Propeller Overspeed

RPM above 2700 and not responding to prop lever

Throttle	**REDUCE**
Propeller	**LOW RPM (full aft)**
Airspeed	**REDUCE (pitch up) to bring RPM within limits**
Oil pressure	**CHECK**
Power	**SET minimum that keeps RPM at or below 2700**
Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical.	**NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

Throttle (memory)	**CLOSED**
Brakes (memory)	**AS REQUIRED – stop straight ahead**
Mixture	**IDLE CUTOFF**
Fuel selector	**OFF**
Ignition	**OFF**
Battery & alternator switches	**OFF**

Engine Failure After Takeoff

Airborne at low altitude

LAND IMMEDIATELY

Airspeed (memory)	**85–90 KIAS (110 KIAS gear & flaps up)**

▶ Runway remaining to land on?

Yes:

Landing gear	**DOWN (if retracted)**
Flaps	**AS REQUIRED**
Land	**STRAIGHT AHEAD**

No – land ahead off-field:

Landing gear	**DOWN on firm surface / UP on rough terrain or water**
Flaps	**DOWN when landing assured**
Fuel selector	**OFF**
Mixture	**IDLE CUTOFF**
Ignition	**OFF**
Battery & alternator switches	**OFF (after gear position set)**

Land straight ahead with only small changes of direction. Do not attempt a turn back.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	**110 KIAS (best glide, gear & flaps up)**
Fuel selector (memory)	**SWITCH TANKS (feel for detent)**
Auxiliary fuel pump (memory)	**HI, then LO if fuel flow restored**
Mixture (memory)	**FULL RICH, then lean as required**
Propeller (memory)	**LOW RPM (full aft) to extend glide**

▶ Restart attempt practicable?

Yes – restart:

Ignition	**BOTH – check**
Throttle	**1/4 OPEN**
Starter	**ENGAGE if propeller has stopped**

▶ Engine restarted?

Yes:

Power	**AS REQUIRED**
Auxiliary fuel pump	**LO if required**
Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel system.	**NOTED**

No:

Continue with forced landing.

No – forced landing:

Forced landing without power:

Seats / belts / harnesses	**SECURE**
Landing gear	**DOWN on firm surface (below 154 KIAS) / UP on rough terrain or water**
Flaps	**DOWN when landing assured**
Airspeed	**80 KIAS on final**
Fuel selector	**OFF**
Mixture	**IDLE CUTOFF**
Ignition	**OFF**
Battery & alternator switches	**OFF (after gear and flaps set)**
Cabin door	**UNLATCH before touchdown**

Engine Fire in Flight

LAND IMMEDIATELY

Fuel selector (memory)	**OFF**
Mixture (memory)	**IDLE CUTOFF**
Auxiliary fuel pump (memory)	**OFF**
Throttle (memory)	**CLOSED**
Cabin heat / defroster / vents	**OFF / CLOSED**
Airspeed	**INCREASE (up to 154 KIAS) to help extinguish fire**
Landing gear	**DOWN electrically before switching battery off (or hand-crank)**
Forced landing	**EXECUTE (do not attempt restart)**

Engine Fire During Start

Starter (memory) ..... **CONTINUE CRANKING**

▶ Engine starts?

Yes:

- Throttle ..... **1700 RPM for a few minutes**
- Engine ..... **SHUT DOWN and inspect for damage**

No:

- Mixture ..... **IDLE CUTOFF**
- Throttle ..... **FULL OPEN**
- Auxiliary fuel pump ..... **OFF**
- Fuel selector ..... **OFF**
- Starter ..... **CONTINUE CRANKING to draw fire into engine**
- Ignition ..... **OFF**
- Battery & alternator switches ..... **OFF**
- Fire extinguisher ..... **OBTAIN / USE**
- Aircraft ..... **EVACUATE**

Cabin / Electrical Fire

LAND AS SOON AS POSSIBLE

Battery & alternator switches (memory) ..... **OFF**

Avionics master (memory) ..... **OFF**

Cabin heat / vents ..... **CLOSED**

Fire extinguisher ..... **ACTIVATE (then ventilate cabin)**

▶ Fire out and electrical power required?

Yes:

- Circuit breakers ..... **CHECK – do not reset faulted**
- Battery switch ..... **ON**
- Alternator / avionics / equipment ..... **ON one at a time, check for smoke**
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

No:

Landing gear and flaps are electric: plan a hand-crank gear extension (see Emergency Gear Extension) and a flaps-up landing.

- Deferred: Electrical power off – manual gear extension, flaps-up landing, no landing light. ..... **NOTED**

Emergency Descent

Throttle (memory) ..... **CLOSED**

Propeller (memory) ..... **HIGH RPM**

Landing gear (memory) ..... **DOWN (below 154 KIAS)**

Flaps (memory) ..... **APPROACH (below 152 KIAS)**

Airspeed (memory) ..... **154 KIAS (smooth air) / 141 KIAS (turbulence)**

Mixture ..... **ENRICH as required**

Emergency Gear Extension (Hand Crank)

Gear does not extend electrically

Airspeed	**100 KIAS or less**
Landing gear motor circuit breaker	**PULL**
Landing gear switch	**DOWN**
Hand crank	**REMOVE from stowage and ENGAGE (floor, behind / between front seats)**
Hand crank	**TURN COUNTERCLOCKWISE to the stop (approx 50 turns)**
Landing gear indicator	**GREEN – gear down**
Hand crank	**DISENGAGE & STOW**

Do not reset the gear motor circuit breaker or move the gear switch in flight after a manual extension. Leave the gear down.

▶ Green light obtained?

Yes:

Deferred: Gear extended manually – do not retract; gear system inspection required. ..... **NOTED**

No:

Continue to Gear-Up / Partial Gear Landing procedure.

Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

Runway	**LONGEST HARD SURFACE available, emergency services advised**
Passenger briefing	**COMPLETE (belts tight, brace, exits)**
Landing gear	**UP (or as it is – do not cycle)**
Flaps	**DOWN**
Approach	**NORMAL, 75–80 KIAS**
Auxiliary fuel pump	**OFF**
Cabin door	**UNLATCHED**

On short final:

Mixture	**IDLE CUTOFF**
Fuel selector	**OFF**
Ignition	**OFF**
Battery & alternator switches	**OFF**
Touchdown	**MINIMUM SPEED, slightly nose high, wings level**
Aircraft	**EVACUATE once stopped**

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist version 1.8 (build c9e52546) · 2026-11-07